

Article

Correlation-Based Single-Phase Heat Transfer Assessment of Binary HFE/Ethyl Acetate Mixtures in Minichannels

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Abstract

This work presents a correlation-based framework for comparative assessment of single-phase forced convection of binary hydrofluoroether/ethyl acetate (HFE/EA) mixtures in rectangular minichannels. Density, kinematic viscosity, and thermal conductivity were measured at 293.1, 313.1, and 328.1 K for selected compositions of HFE-7100/EA, HFE-7300/EA, and HFE-73DE/EA. Because several DSC-derived mixture-specific heat values were not sufficiently reliable for direct use, the mixture-specific heat capacity was estimated from literature-supported pure-component values using an ideal-mixture, mass-fraction-weighted approximation and used only for evaluation of the Prandtl number. Heat transfer was then assessed using the Sieder–Tate correlation for laminar thermally developing flow in two representative minichannel geometries. The highest predicted values for the HFE-7100/EA and HFE-7300/EA families were obtained for the most EA-rich retained compositions, whereas in the retained HFE-73DE/EA subset, the 50/50 mixture performed best because of its higher thermal conductivity. Validation against an experimental dataset for pure HFE-7100 in the short module showed systematic overprediction, with a mean relative difference of −13.44% and a MAPE of 15.6%. The calculated values should, therefore, be used for relative comparison rather than treated as unbiased absolute predictions.

Keywords: binary mixtures; thermophysical properties; dielectric fluids; minichannel; heat transfer coefficient; Nusselt number; single-phase forced convection

Academic Editor: Marco Marengo

Received: 6 March 2026

Revised: 27 April 2026

Accepted: 5 May 2026

Published: 9 May 2026

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1. Introduction

Heat transfer devices are used in a wide range of technologies, including refrigeration, HVAC systems, thermal management of high-power electronics, cryogenic equipment, photovoltaic systems, and industrial processes. Their operating efficiency is largely governed by the properties of the selected coolant or process fluid. In practice, the fluid must be compatible with the required temperature and pressure levels, expected flow conditions, construction materials, and safety requirements. Fluorinated liquids, including hydrofluoroethers (HFEs), perfluoropolyethers (PFPEs), and Fluorinert-type fluids (FCs), are often considered for such applications because they combine dielectric behaviour with good chemical inertness and thermal resistance. Their normal boiling points cover a broad temperature interval, approximately from 300 K to more than 500 K, which

makes them suitable for the thermal control of electronic and optoelectronic devices. A drawback of these fluids is their relatively poor heat conduction and, in many cases, high price. For this reason, increasing attention is being paid to liquid blends whose properties can be adjusted for a given thermal process. In particular, binary systems composed of fluorinated liquids and organic solvents, or two fluorinated components, offer a practical route to modify boiling point, density, viscosity, and specific heat while preserving acceptable material compatibility and ease of circulation in typical pumping systems.

The work presented in [1] proposes the use of differential scanning calorimetry (DSC) for acquiring atmospheric-pressure boiling-point and vapor–liquid equilibrium data for binary liquid mixtures. The approach is especially relevant to mixtures whose components exhibit large differences in volatility, since such systems are susceptible to pre-evaporation effects during measurement. These errors were limited by optimising the sample size and heating programme. The methodology was first assessed for several model mixtures and was then used to investigate the previously undocumented carvacrol–DMSO binary system. The experimental boiling line was correlated with the NRTL thermodynamic model, which confirmed significant non-ideal behaviour and suggested the presence of an azeotropic point.

Recent studies show a clear shift towards computational and machine-learning tools for predicting properties of HFE-based and related fluid mixtures. For example, Aminian et al. [2] combined *ab initio* calculations with the Peng–Robinson equation of state to evaluate ideal-gas heat capacities and critical properties of HFE-7000–7500 compounds. In another contribution, Urata et al. [3] applied artificial neural networks to estimate vapour–liquid equilibrium behaviour in HFE-containing mixtures. Similar data-oriented strategies have been extended to other refrigerant and hydrocarbon systems. Wu et al. [4] proposed a machine-learning-assisted group-contribution approach for predicting critical temperatures of binary refrigerant mixtures from molecular information, while Soria-Lopez et al. [5] reported improved viscosity predictions for binary alkane mixtures compared with traditional mixing rules, owing to the ability of ML models to capture nonlinear dependence on composition.

Reliable high-pressure property data are available for several HFE-containing binary systems. For the HFE-7200 + 2-propanol mixture, Muñoz-Rujas et al. [6] measured thermophysical properties between 293.15 and 393.15 K and at pressures from 0.1 to 140 MPa. The authors represented the density data using a Tait-type equation, calculated related derivative properties, and supplemented the analysis with speed-of-sound results at 0.1 MPa. In another study [7], analogous density and sound-speed measurements were reported for HFE-7500 + diisopropyl ether mixtures up to 100 MPa. Thermodynamic properties of hydrofluoroether systems with selected organic solvents were further investigated by Ogawa et al. [8], who observed positive excess molar volumes and excess enthalpies dependent on composition.

From an engineering viewpoint, zeotropic mixtures offer promising opportunities for improving heat transfer performance. In pulsating heat pipes, Xu et al. [9] observed that partially immiscible zeotropic fluids can form dispersed emulsion structures, leading to enhanced heat transfer. For condensation processes, Eissa et al. [10] formulated a generalised non-equilibrium heat transfer model for binary zeotropic mixtures. Their approach includes vapour-side mass diffusion and interfacial temperature effects, and its validation against 871 experimental data points showed strong predictive capability, with 92% of the results lying within a $\pm 30\%$ deviation band. This makes the model relevant for compact heat-exchanger analysis and design.

Tagliaferri et al. [11] showed that, for CFD of bubbling fluidized bidisperse mixtures, a first-order upwind discretization can introduce strong numerical diffusion that suppresses bubble holdup and mixing, whereas a second-order upwind scheme yields more

realistic bubble dynamics and correctly predicts the transition to a fully mixed state; the restitution coefficient has a minor influence except near the quasi-ideal value $e \approx 0.99$, and explicit vs implicit time integration mainly affects the transient duration rather than the pseudo-steady behaviour.

Additional studies have addressed related features of binary liquid systems, such as intermolecular effects, equilibrium characteristics, and behaviour at phase boundaries. Verma et al. [12] used density, ultrasonic velocity, and refractive-index measurements to show pronounced donor–acceptor interactions in ether/hydrocarbon mixtures. Tsai et al. [13], in turn, evaluated the ability of molecular dynamics simulations to predict transport properties in alcohol/water systems and indicated which force-field combinations most accurately describe diffusion coefficients and density variations.

In previous work [14], the authors combined thermophysical property measurements with validated conjugate CFD to assess HFE-73DE/ethyl acetate mixtures for laminar minichannel cooling and to identify compositions offering a heat transfer/pressure-drop compromise. In that study, the thermophysical properties of the working fluids were determined experimentally at the Faculty of Civil Engineering, Mechanics and Petrochemistry, Warsaw University of Technology in Płock, Poland. Liquid density was measured with a Mettler Toledo Densito densimeter in accordance with ASTM D1250 [15], while kinematic viscosity over the investigated temperature range was determined using a Ubbelohde capillary viscometer thermostated in water baths according to EN ISO 3104 [16]. The initial boiling point was determined by atmospheric distillation following ASTM D86 [17]. Thermal measurements were performed by differential scanning calorimetry (DSC) using a Netzsch Maia 200 F3 instrument, with an empty crucible as the reference and liquid sample masses of approximately 60–70 mg [18,19].

These measurements provided the experimental property basis for the present comparative assessment. The mixtures were prepared in the laboratory from commercially available components rather than purchased as ready-made formulations. Because this study aims at a correlation-based comparison under identical hydraulic assumptions, directly measured temperature-dependent properties were treated as the primary input data wherever their consistency was confirmed. This is particularly important for dielectric multicomponent systems, for which simple mixing rules may be insufficient to represent the actual behaviour over the full temperature and composition range. The present work uses that earlier experimental property base as an input layer for a simplified correlation-based comparison, rather than repeating the full conjugate CFD framework reported previously [14]. However, because several DSC-derived mixture-specific heat values were found to be insufficiently reliable for direct use in the final calculations, the specific heat capacity of the mixtures was estimated from pure-component data using an ideal-mixture, mass-fraction-weighted mixing rule.

For each mixture series (HFE-7100/EA, HFE-7300/EA, and HFE-73DE/EA), mixtures were considered at selected mass fractions (first component/second component) and characterised at 293.1, 313.1, and 328.1 K under near-atmospheric pressure. The dataset used in the final comparative analysis includes experimentally determined density, kinematic viscosity, and thermal conductivity, while specific heat was estimated from pure-component mass fractions for calculation of the Prandtl number. This article presents a transparent, correlation-based workflow for converting the measured thermophysical properties into comparable single-phase forced-convection heat transfer characteristics for rectangular minichannels. The novelty lies in combining a dedicated property dataset for HFE/EA mixtures with a consistent minichannel assessment framework applied to two representative module geometries and in anchoring the assessment with an independent validation dataset for pure HFE-7100 obtained in the short multi-minichannel module using IR thermography and an energy-balance data reduction.

2. Characteristics of Pure Liquids and Selected Binary Mixtures

The present study considers three binary mixture series based on hydrofluoroethers (HFEs) and an organic ester: 3M™ Novec™ HFE-7100/ Ethyl acetate EA, 3M™ Novec™ HFE-7300/Ethyl acetate EA, and 3M™ Novec™ HFE-73DE/ Ethyl acetate EA. These liquids were selected because they are electrically insulating, chemically stable, and relevant to thermal-management and compact heat-exchanger applications, while the ester co-solvent enables tunability of density, viscosity, and boiling behaviour.

Table 1 provides selected reference properties of the base fluids (reported at 298 K unless stated otherwise). For HFE-7100 [20], HFE-7300 [21], and HFE-73DE [22], the listed densities, viscosities, and boiling temperatures illustrate the range of base-fluid properties relevant to minichannel applications. Ethyl acetate [23] is included as the organic co-solvent used to modify the density, viscosity, and boiling behaviour of the analysed HFE/EA mixtures.

Table 1. Selected reference properties of the base fluids [20–23].

Physical Properties	HFE-7100	HFE-7300	HFE-73DE	EA
Density, kg/m ³	1520	1660	1280	902
Viscosity, mm ² /s	0.401	0.71	0.403	0.463
Boiling temperature, K	334.1	371.1	321.1	350.2

The HFE-7100/EA, HFE-7300/EA, and HFE-73DE/EA systems were considered at mass fractions of 10/90, 25/75, 50/50, and 75/25 and at three temperature levels. The 10/90 composition was available only for HFE-7100/EA and HFE-73DE/EA. The 75/25 HFE-73DE/EA case was excluded from the final dataset, and all remaining compositions were used as input to the correlation-based calculations reported in the following tables.

3. Experimental Rig and Minichannel Modules

3.1. General Information

The present study comprises (i) a correlation-based assessment of selected binary mixtures using experimentally measured mixture properties, performed for two minichannel modules—a long reference module (Setup #1) and a short multi-minichannel module (Setup #2), and (ii) an experimental validation dataset acquired in the short module (Setup #2) using a pure reference fluid (HFE-7100). For the flow of either mixtures or pure liquids in the single minichannel (Setup #1) or in the central minichannel of the parallel-channel group (Setup #2), laminar single-phase forced convection was assumed.

Setup #1, incorporating the long minichannel, had been investigated in the authors' previous experiments using liquid crystal thermography to obtain the outer-wall temperature distribution of the heated foil. The corresponding data were not considered in the present study [24].

The configuration with a short minichannel module (Setup #2) was used in recent experimental studies by the authors, in which infrared thermography was employed to measure the temperature of the outer surface of the heated foil [25]. These IR-based data were selected for the present analysis because they provide a more complete and reliable wall-temperature field. In the liquid-crystal thermography measurements, portions of the foil surface could not be evaluated when the local temperature fell outside the active temperature range of the liquid-crystal coating; therefore, this technique was not adopted for the validation dataset used here.

Uncertainty estimates for the principal measured variables were reported in [24,25]. Since the heat transfer coefficient is calculated from temperature-dependent quantities, its uncertainty is largely governed by the precision of the temperature measurements.

It should also be noted that the mixtures addressed in this article were not tested experimentally in the heat transfer rig. Therefore, the heat transfer coefficients and Nusselt numbers reported for the mixtures are calculated estimates obtained from the measured thermophysical properties and a selected literature correlation, applied to the geometries and operating assumptions defined in this work.

3.2. Experimental Rig with a Minichannel Module

The validation experiments were performed in single-phase forced convection. The working fluid was circulated through the test section at controlled inlet temperature and mass flow rate, while the heat input was supplied electrically at the heated wall of the minichannel module (Setup #2). Inlet and outlet temperatures and pressures were monitored continuously, while infrared (IR) thermography was used to capture the temperature distribution over the external side of the heater foil, yielding spatially resolved thermal data. The validation dataset for the reference fluid (HFE-7100) was collected using Setup #2.

A schematic of the experimental loop and the interchangeable test sections is shown in Figure 1.

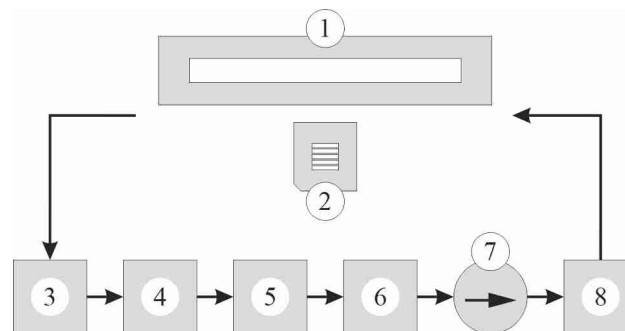


Figure 1. Schematic diagram of the experimental facility: 1—Setup #1 (long module with the minichannel); 2—Setup #2 (short module with the 5 minichannels); #3—condenser; #4—pressure-control unit; #5—deaerator; #6—filter; #7—pump; #8—mass flow meter.

In the experimental loop, the working fluid was pumped to a pressure-control vessel, which also acted as an expansion chamber. The vessel was divided into two zones: one filled with liquid and the other with compressed air, allowing the required system pressure to be maintained. After leaving this unit, the fluid flowed through a preheater, where it was brought to the prescribed inlet temperature, usually slightly below the saturation point, before entering the test section. Further details of the experimental facility are given in [24,25]. All tests were performed under steady-state conditions for several values of electrical power delivered to the heater, which produced different imposed heat-flux levels.

3.3. Reference Minichannel Modules

Both modules comprise minichannels with a depth of 1 mm. In each minichannel, the heated wall was made of Haynes-230 alloy foil, approximately 0.1 mm in thickness, the opposite wall was a glass window, and the side walls were made of PTFE. This work focuses on the short module with five parallel minichannels (Setup #2), which is schematically shown in Figure 2.

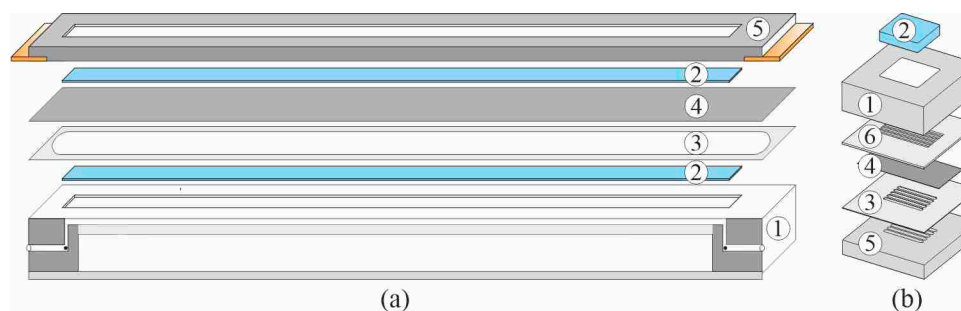


Figure 2. Schematic diagrams of (a) the long module with a single minichannel (Setup #1), (b) the short module with five parallel minichannels (Setup #2), 1—body, 2—glass plate, 3—PTFE mini-channel insert, 4—heated foil, 5—cover plate, and 6—PTFE electrically insulating insert.

The principal geometrical dimensions of both modules are summarised in Table 2. Setup #1 employs a single minichannel with a length of 0.36 m, whereas Setup #2 uses five parallel minichannels, each 0.043 m long; in both modules, the minichannel depth is 1 mm.

Table 2. Geometrical characteristics of the test modules.

	Setup #1	Setup #2
Number of minichannels (-)	1	5
Minichannel length L (m)	0.36	0.043
Minichannel width w (m)	0.004	0.006
Minichannel depth s (m)	0.001	0.001

4. Calculations and Correlation-Based Assessment Method

To compare selected mixtures of HFE-7100, HFE-7300, HFE-73DE, and ethyl acetate (EA) under identical hydraulic conditions and a fixed operating point, a correlation-based assessment approach was adopted for single-phase forced convection in a reference rectangular minichannel. The assessment geometry and operating point were fixed as follows: channel width $w = 0.004$ m, channel depth $s = 0.001$ m, length $L = 0.36$ m, hydraulic diameter $D_h = 1.6$ mm, and mass flux $G = 200$ kg/(m²·s).

All thermophysical property measurements for the mixtures were carried out at the Faculty of Civil Engineering, Mechanics and Petrochemistry, Warsaw University of Technology in Płock (Poland). The dataset was collected at three temperature levels (293.1, 313.1, and 328.1 K). In the final comparative calculations, experimentally determined density, kinematic viscosity, and thermal conductivity were used directly. However, several DSC-derived mixture-specific heat values were found to be insufficiently reliable for direct use in the correlation-based calculations, mainly due to volatility-related measurement artefacts and the lack of a repeat verification campaign at the revision stage. Therefore, in the present manuscript, specific heat was not taken directly from the disputed mixture DSC results. Instead, c_p values for the mixtures were estimated from pure-component mass fractions and used only for evaluation of the Prandtl number within the comparative assessment framework. Details of the original property measurement procedures are given in [14].

For each mixture and temperature, the dynamic viscosity was computed from density and kinematic viscosity. The Reynolds and Prandtl numbers were then calculated according to the following relations:

$$Re = \frac{G \cdot D_h}{\mu} \quad (1)$$

$$Pr = \frac{\mu c_p}{k} \quad (2)$$

where D_h is the hydraulic diameter of the minichannel, G is mass flux, μ is the dynamic viscosity, c_p is the specific heat, and k is the thermal conductivity of the fluid.

The average Nusselt number Nu_{corr} was estimated using the classical Sieder–Tate-type correlation [26] for laminar internal flow with thermal development effects:

$$Nu_{corr} = 1.86 \left(Re \cdot Pr \cdot \frac{D_h}{L} \right)^{\frac{1}{3}} \left(\frac{\mu}{\mu_w} \right)^{0.14} \quad (3)$$

where L is the heated length of the minichannel.

The viscosity ratio is the Sieder–Tate correction accounting for property variations between the bulk fluid and the wall: μ is the dynamic viscosity evaluated at the bulk (mean) fluid temperature, while μ_w is the dynamic viscosity evaluated at the wall temperature. In the present calculations, this viscosity-ratio correction was not applied, and $(\mu/\mu_w)^{0.14}$ was set to unity. This choice keeps the workflow uniform across all mixtures and temperature levels and avoids introducing an additional wall-property model not supported by direct mixture data. Its effect is modest unless the wall-to-bulk viscosity contrast is strong: for $\mu/\mu_w = 0.8, 1.2, 1.5$, and 2.0 , the correction factor equals $0.969, 1.026, 1.058$, and 1.102 , respectively. Therefore, omission of the term should be interpreted as acceptable for first-stage comparative assessment, but not as a substitute for a detailed local-property treatment. The use of the hydraulic diameter to extend tube correlations to non-circular ducts is standard practice in preliminary internal-flow heat transfer estimates, while detailed aspect-ratio-dependent solutions for fully developed laminar duct flow can be found in dedicated duct-flow sources [27]. Accordingly, the Sieder–Tate correlation is used here as a uniform comparative metric across all cases, not as a geometry-specific predictive model for rectangular minichannels.

Finally, the heat transfer coefficient was obtained from the definition of the Nusselt number Nu ($Nu = h \cdot D_h / k$, where h is the heat transfer coefficient). Using the Sieder–Tate correlation to compute Nu_{corr} , the corresponding coefficient h_{corr} is

$$h_{corr} = \frac{Nu_{corr} \cdot k}{D_h} \quad (4)$$

where k is the thermal conductivity evaluated at the bulk (reference) fluid temperature.

The assessment is restricted to single-phase forced convection with no phase change, no conjugate heat conduction in the wall, and no pressure-drop constraint included in the ranking criterion. Properties are evaluated at discrete temperatures (293.1/313.1/328.1 K) and assumed uniform for each case. The adopted correlation targets laminar flow with thermal development along the heated length; therefore, the computed Nu and h should be interpreted as relative heat transfer metrics under the specified geometry and operating point, suitable for experimental validation.

Because the present framework is intended for comparative assessment rather than high-fidelity local prediction, the adopted property treatment was kept deliberately simple and internally consistent across all cases. Density, kinematic viscosity, and thermal conductivity were taken directly from the experimentally determined mixture dataset at each temperature level, whereas mixture-specific heat capacity was estimated from pure-component data only for evaluation of the Prandtl number. This approach preserves the measured transport-property trends while avoiding direct use of disputed DSC-derived mixture c_p values in the final calculations.

5. Results and Analysis

5.1. Heat Transfer Assessment Outputs for Investigated Mixtures

This section presents the measured thermophysical property trends and the resulting heat transfer assessment outputs for the investigated mixtures.

For each mixture series (HFE-7100/EA, HFE-7300/EA, and HFE-73DE/EA), mixtures were considered at selected mass fractions (first component/second component) and characterised at 293.1, 313.1, and 328.1 K under near-atmospheric pressure. The dataset used in the final comparative analysis includes experimentally determined density, kinematic viscosity, and thermal conductivity, while specific heat was estimated from pure-component mass fractions. These properties were used to evaluate the Reynolds and Prandtl numbers. For each mixture and temperature, dynamic viscosity was calculated from the experimentally determined density and kinematic viscosity and then used in the Reynolds- and Prandtl-number calculations.

The specific heat capacity of each mixture was estimated using an ideal-mixture, mass-fraction-weighted mixing approximation, according to the following relation [28]:

$$c_{p,mix}(T) = \sum w_i \cdot c_{p,i}(T) \quad (5)$$

where $c_{p,mix}(T)$ is the estimated isobaric specific heat capacity of the mixture at temperature T , w_i is the mass fraction of component i , and $c_{p,i}(T)$ is the isobaric specific heat capacity of pure component i at the same temperature. This approximation corresponds to neglecting the excess heat-capacity contribution of the mixture.

Pure-component c_p values were assembled as follows. For ethyl acetate, liquid c_p data from the NIST Chemistry WebBook [23] were used to assign near-ambient values at 293.1, 313.1, and 328.1 K. For HFE-7300, ambient-pressure liquid c_p data from the thermodynamic study of Kocian et al. [29] were interpolated to the three target temperatures. For HFE-7100, a literature-backed reference value of 1183 J/(kg·K) at 25 °C [30] was retained as a narrow-range approximation across the analysed temperatures. For HFE-73DE, no reliable public temperature-dependent $c_p(T)$ dataset was identified; therefore, the reference value of 1201 J/(kg·K), reported in [31], was used as a constant approximation. Because c_p enters the adopted screening only through Pr and then through the one-third power in Equation (3), the resulting sensitivity of Nu_{corr} and h_{corr} to residual c_p uncertainty is limited. These assumptions should nevertheless be treated as a limitation of the present comparative screening framework.

Table 3 summarises the physical properties of the mixtures at 293.1 K, Table 4 shows the data at 313.1 K, whereas Table 5 concerns data at 328.1 K. Within the analysed composition ranges, density, viscosity, thermal conductivity, and specific heat vary systematically with both temperature and composition, providing a consistent basis for comparative thermal-hydraulic assessment.

Table 3. Summary of experimentally determined properties (ρ , ν , k), estimated specific heat capacity c_p obtained from the mass-fraction-weighted mixing rule, and assessment outputs for all selected mixtures and compositions retained for analysis at $T = 293.1$ K (reference Setup #1: $w = 4$ mm, $s = 1$ mm, $L = 0.36$ m, $D_h = 1.6$ mm, $G = 200$ kg/(m²·s)).

Mixture	Mass Fraction (%)	ρ (kg/m ³)	μ (mPa·s)	k (W/(m·K))	c_p (J/(kg·K))	Re (–)	Nu_{corr} (–)	h_{corr} (W/(m ² ·K))
HFE-73DE/EA	10/90	931	0.4731	0.165	1848.5	676.4	4.68	482.7
HFE-73DE/EA	25/75	969	0.4877	0.189	1740.6	656.1	4.38	517.9
HFE-73DE/EA	50/50	1035	0.4839	0.207	1560.8	661.3	4.10	530.6
HFE-7100/EA	10/90	940	0.5194	0.125	1846.8	616.1	5.13	401.0
HFE-7100/EA	25/75	1017	0.5039	0.116	1736.1	635.0	5.15	373.7
HFE-7100/EA	50/50	1111	0.5341	0.105	1551.8	599.1	5.13	336.9
HFE-7100/EA	75/25	1216	0.5787	0.089	1367.4	553.0	5.20	289.2
HFE-7300/EA	25/75	1006	0.5005	0.133	1719.3	639.4	4.91	408.1
HFE-7300/EA	50/50	1154	0.578	0.101	1518.0	553.6	5.16	325.9
HFE-7300/EA	75/25	1348	0.741	0.087	1316.8	431.8	5.17	281.3

Table 4. Summary of experimentally determined properties (ρ , ν , k), estimated specific heat capacity obtained from the mass-fraction-weighted mixing rule, and assessment outputs for all selected mixtures and compositions retained for analysis at $T = 313.1$ K (reference Setup #1: $w = 4$ mm, $s = 1$ mm, $L = 0.36$ m, $D_h = 1.6$ mm, $G = 200$ kg/(m²·s)).

Mixture	Mass Fraction (%)	ρ (kg/m ³)	μ (mPa·s)	k (W/(m·K))	c_p (J/(kg·K))	Re (–)	Nu_{corr} (–)	h_{corr} (W/(m ² ·K))
HFE-73DE/EA	10/90	902.1	0.3804	0.155	1852.7	841.2	4.78	463.3
HFE-73DE/EA	25/75	951.7	0.399	0.177	1744.1	802.0	4.48	496.1
HFE-73DE/EA	50/50	933.0	0.3613	0.194	1563.0	885.7	4.19	508.4
HFE-7100/EA	10/90	888.0	0.3719	0.123	1850.9	860.4	5.16	397.0
HFE-7100/EA	25/75	785.2	0.3239	0.111	1739.6	988.0	5.23	363.1
HFE-7100/EA	50/50	1044.0	0.4143	0.1	1554.0	772.4	5.22	326.2
HFE-7100/EA	75/25	1042.3	0.4158	0.082	1368.5	769.6	5.35	274.0
HFE-7300/EA	25/75	1067.4	0.4475	0.107	1730.2	715.1	5.29	353.7
HFE-7300/EA	50/50	1138.0	0.4603	0.095	1535.3	695.2	5.29	314.0
HFE-7300/EA	75/25	1470.8	0.6413	0.08	1340.4	499.0	5.35	267.6

Table 5. Summary of experimentally determined properties (ρ , ν , k), estimated specific heat capacity obtained from the mass-fraction-weighted mixing rule, and assessment outputs for all selected mixtures and compositions retained for analysis at $T = 328.1$ K. (reference Setup #1: $w = 4$ mm, $s = 1$ mm, $L = 0.36$ m, $D_h = 1.6$ mm, $G = 200$ kg/(m²·s)).

Mixture	Mass Fraction (%)	ρ (kg/m ³)	μ (mPa·s)	k (W/(m·K))	c_p (J/(kg·K))	Re (–)	Nu_{corr} (–)	h_{corr} (W/(m ² ·K))
HFE-73DE/EA	10/90	894.4	0.3292	0.148	1858.8	972.1	4.86	449.7
HFE-73DE/EA	25/75	951.6	0.3543	0.169	1749.2	903.2	4.56	481.5
HFE-73DE/EA	50/50	916.3	0.3478	0.185	1566.5	920.1	4.26	492.9
HFE-7100/EA	10/90	888.0	0.327	0.12	1857.0	978.6	5.21	390.9
HFE-7100/EA	25/75	766.3	0.2802	0.11	1744.7	1142.0	5.26	361.3
HFE-7100/EA	50/50	1030.0	0.3594	0.097	1557.5	890.4	5.28	319.9
HFE-7100/EA	75/25	1022.0	0.3431	0.081	1370.2	932.7	5.37	271.8
HFE-7300/EA	25/75	1084.0	0.4036	0.097	1740.9	792.9	5.48	332.0
HFE-7300/EA	50/50	1107.2	0.3906	0.093	1550.0	819.3	5.34	310.6
HFE-7300/EA	75/25	1458.6	0.5479	0.073	1359.0	584.0	5.54	252.9

5.2. Thermophysical Property Trends

Across all mixture series, density decreases with temperature and changes systematically with composition. Kinematic viscosity also decreases with temperature, which increases the Reynolds number under the fixed mass-flux conditions adopted here. However, because the viscosity-ratio correction term was set to unity, the adopted correlation yields $Re \cdot Pr$ independent of viscosity for fixed geometry and mass flux. As a result, $Nu_{corr} \propto (c_p/k)^{1/3}$, while $h_{corr} \propto c_p^{1/3} k^{2/3}$. Under these assumptions, thermal conductivity is the dominant property controlling the predicted h_{corr} levels, while viscosity primarily affects the hydraulic interpretation and verification of the laminar-flow range.

5.3. Correlation-Based Assessment Results for Setup #1 (Long Test Section)

The long-channel assessment (Tables 3–5), together with Figure 3, shows clear differences between mixture families within the compositions retained for analysis. Under the fixed reference operating point, EA-rich compositions generally yield higher correlation-based heat transfer coefficients.

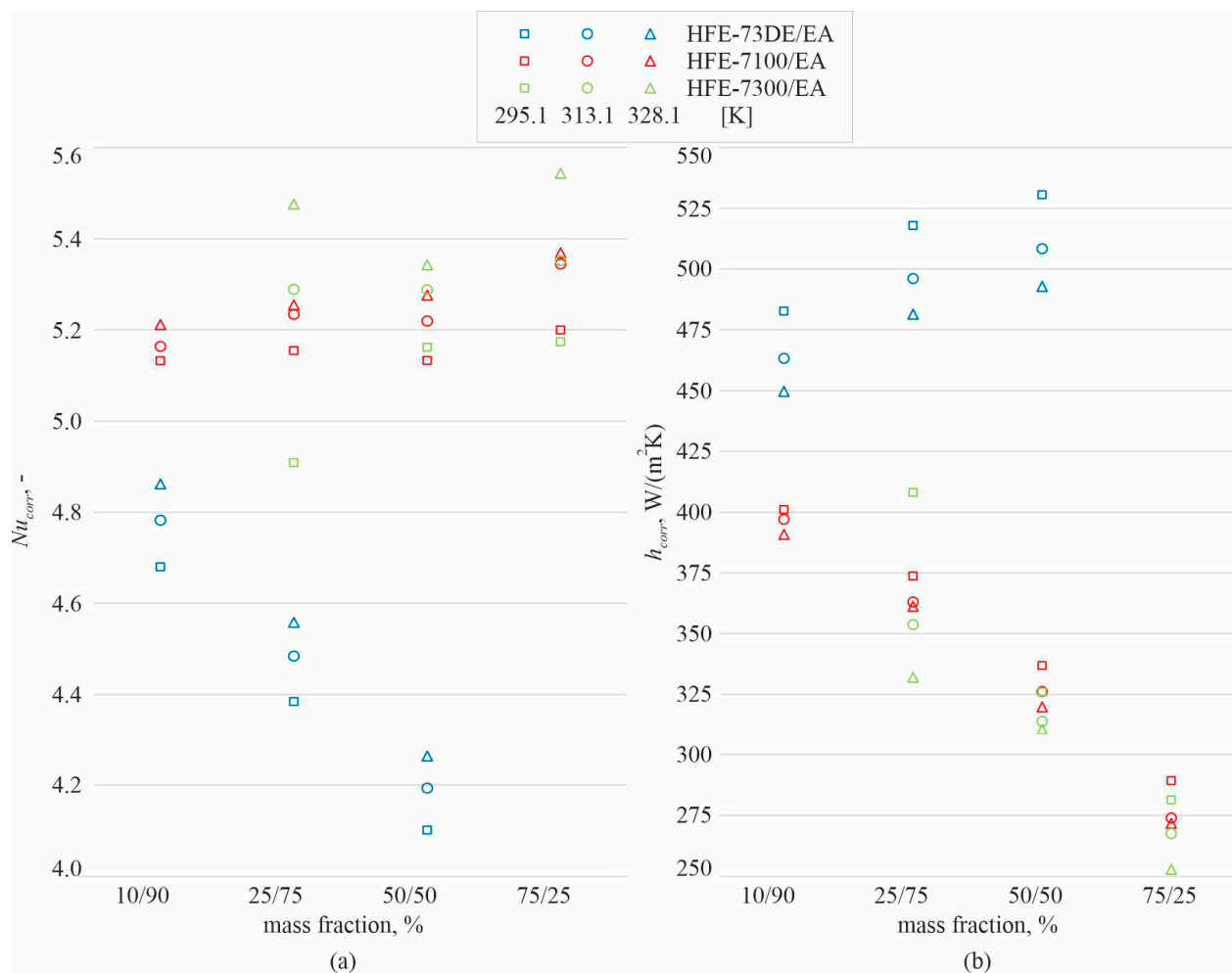


Figure 3. Correlation-based estimates under the Setup #1 reference conditions: (a) Nusselt number versus mass fraction; (b) heat transfer coefficient versus mass fraction for the binary mixtures.

Under the present assumptions, this trend reflects primarily the effect of thermal conductivity and, to a lesser extent, the contribution of c_p through the corrected Prandtl number treatment. Reynolds number is still reported for hydraulic interpretation, but it is not the dominant factor in the ranking of h_{corr} when the viscosity-ratio correction is set to unity.

For the HFE-7100/EA family, the highest h_{corr} values occur at the most EA-rich composition available (10/90) across the investigated temperature range. For HFE-7300/EA, the highest h_{corr} values occur at 25/75 throughout the analysed range. For HFE-73DE/EA, the comparison is limited to 10/90–50/50 because the 75/25 composition was excluded; within the retained set, the 50/50 mixture yields the highest h_{corr} at all three temperature levels, followed by 25/75 and 10/90. This difference relative to the HFE-7100/EA and HFE-7300/EA families follows from the stronger contribution of thermal conductivity in the adopted correlation-based ranking.

Using the reference geometry and operating point defined in Section 4, Reynolds and Prandtl numbers were computed for each mixture at 293.1, 313.1, and 328.1 K. In the corrected c_p treatment, an increase in temperature does not automatically increase h_{corr} . In most of the retained cases, h_{corr} decreases with temperature because the reduction in thermal conductivity outweighs the relatively weak one-third-power influence of c_p . Figure 3 summarises the correlation-based Nusselt number and heat transfer coefficient as functions of mass fraction under the Setup #1 reference conditions.

5.4. Correlation-Based Assessment Results for Setup #2 (Short Test Section)

For the short-module geometry, the same correlation framework was applied using the module-specific hydraulic diameter and heated length. In contrast to the long-module assessment, the mass flux was set to the value corresponding to the selected convection-only operating point used for validation (Section 6). This ensures that the correlation-based estimates for Setup #2 are evaluated under the same hydraulic conditions as the experimental dataset, enabling a consistent comparison.

For the Setup #2 assessment (Tables 6–8 and Figure 4), the reference conditions were defined by the Setup #2 geometry (five parallel minichannels with $w = 0.006$, $s = 0.001$ m, and $L = 0.043$ m; Table 2). The hydraulic diameter was evaluated accordingly, and mixture properties at each temperature level were treated as constant reference values, in analogy to the long-module calculations. Based on the experimental operating conditions, a fixed mass flux of $G = 372.5$ kg/(m²·s) was adopted.

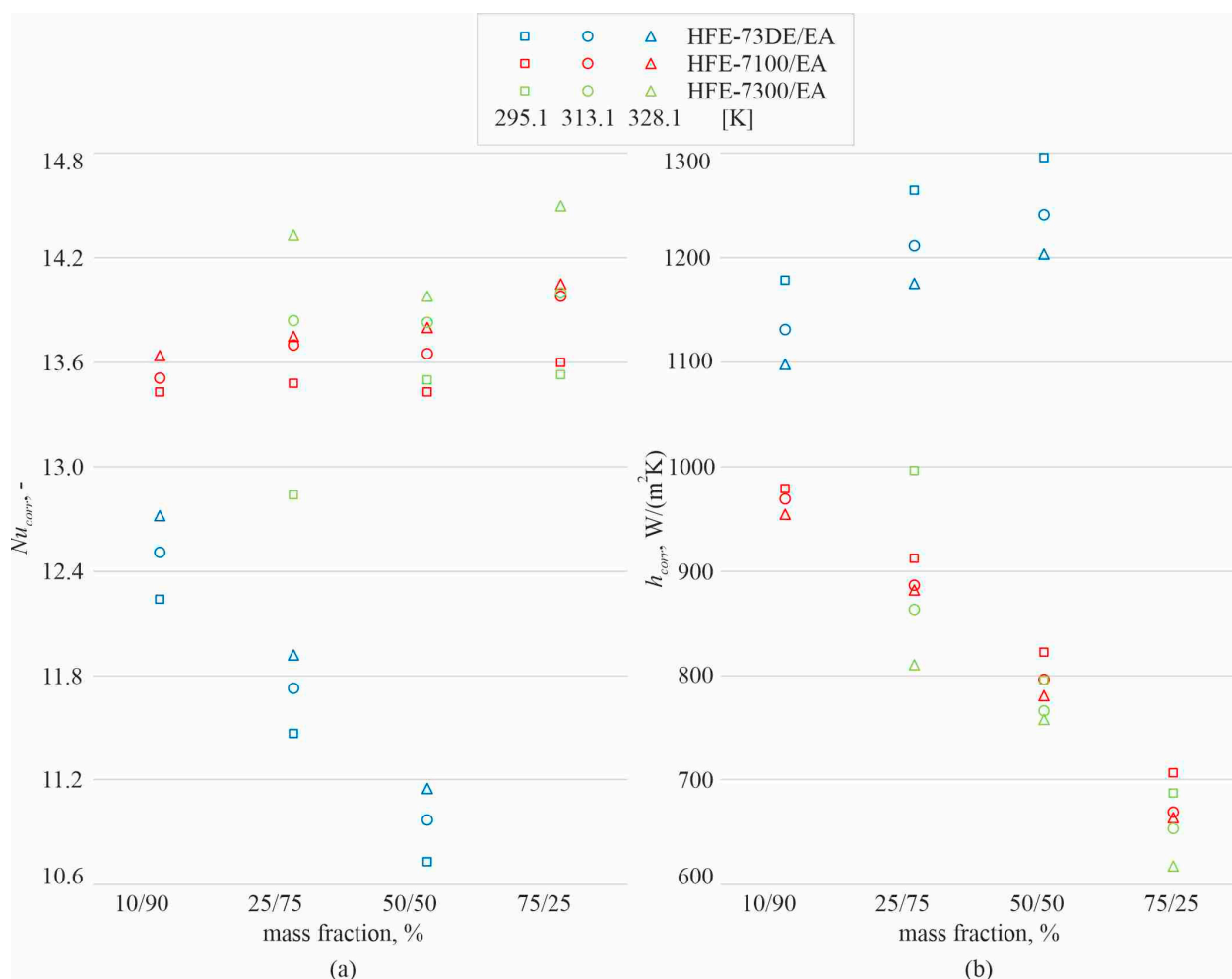


Figure 4. Correlation-based estimates under the Setup #2 reference conditions: (a) Nusselt number versus mass fraction; (b) heat transfer coefficient versus mass fraction for the binary mixtures.

Tables 6–8 compile the correlation-based assessment outputs for the selected mixtures and compositions retained for analysis under the Setup #2 reference conditions at $T = 293.1$, 313.1 , and 328.1 K. Figure 4 illustrates the corresponding Nusselt number and heat transfer coefficient as functions of mass fraction for the binary mixtures under the same Setup #2 reference conditions. The same property interpretation applies here as in Setup #1: the shorter heated length increases the absolute Nu_{corr} and h_{corr} levels, whereas the relative ranking remains governed primarily by the $c_p - k$ combination embedded in the adopted correlation.

The results in Tables 6–8 and Figure 4 show that the mixture ranking is not specific to the long-channel geometry. The same ranking pattern is retained in the short-module geometry, although the absolute Nu_{corr} and h_{corr} levels are higher because of the shorter heated length. Under the present corrected c_p treatment and with the viscosity-ratio term set to unity, the relative ranking is governed primarily by thermal conductivity, with a weaker contribution from c_p through its one-third-power effect. The validation presented in Section 6 shows that the correlation systematically overpredicts h_{corr} for HFE-7100 under the tested conditions; therefore, the reported absolute values should be treated cautiously, whereas the comparative ranking remains the main intended output of the framework.

Table 6. Summary of correlation-based assessment outputs for the selected mixtures and compositions retained for analysis at $T = 293.1$ K under the Setup #2 reference conditions.

Mixture	Mass Fraction (%)	Re (–)	Nu_{corr} (–)	h_{corr} (W/(m ² ·K))
HFE-73DE/EA	10/90	1939.7	12.24	1178.4
HFE-73DE/EA	25/75	1802.4	11.47	1264.5
HFE-73DE/EA	50/50	1836.0	10.73	1295.6
HFE-7100/EA	10/90	1952.9	13.43	979.0
HFE-7100/EA	25/75	2279.3	13.48	912.4
HFE-7100/EA	50/50	1776.9	13.43	822.4
HFE-7100/EA	75/25	1861.0	13.60	706.2
HFE-7300/EA	25/75	1582.3	12.84	996.3
HFE-7300/EA	50/50	1635.0	13.50	795.6
HFE-7300/EA	75/25	1165.5	13.53	686.9

Table 7. Summary of correlation-based assessment outputs for the selected mixtures and compositions retained for analysis at $T = 313.1$ K under the Setup #2 reference conditions.

Mixture	Mass Fraction (%)	Re (–)	Nu_{corr} (–)	h_{corr} (W/(m ² ·K))
HFE-73DE/EA	10/90	1678.5	12.51	1131.2
HFE-73DE/EA	25/75	1600.4	11.73	1211.2
HFE-73DE/EA	50/50	1767.4	10.97	1241.4
HFE-7100/EA	10/90	1717.2	13.51	969.2
HFE-7100/EA	25/75	1971.2	13.70	886.6
HFE-7100/EA	50/50	1541.5	13.65	796.5
HFE-7100/EA	75/25	1535.8	13.98	668.9
HFE-7300/EA	25/75	1426.9	13.84	863.6
HFE-7300/EA	50/50	1387.3	13.83	766.6
HFE-7300/EA	75/25	995.8	14.00	653.4

Table 8. Summary of correlation-based assessment outputs for the selected mixtures and compositions retained for analysis at $T = 328.1$ K under the Setup #2 reference conditions.

Mixture	Mass Fraction (%)	Re (–)	Nu_{corr} (–)	h_{corr} (W/(m ² ·K))
HFE-73DE/EA	10/90	1349.9	12.72	1098.1
HFE-73DE/EA	25/75	1309.4	11.92	1175.5
HFE-73DE/EA	50/50	1319.5	11.15	1203.5
HFE-7100/EA	10/90	1229.5	13.64	954.5
HFE-7100/EA	25/75	1267.1	13.75	882.1
HFE-7100/EA	50/50	1195.5	13.80	781.1
HFE-7100/EA	75/25	1103.5	14.05	663.7
HFE-7300/EA	25/75	1275.8	14.33	810.6
HFE-7300/EA	50/50	1104.9	13.98	758.2
HFE-7300/EA	75/25	861.7	14.50	617.5

6. Validation of the Correlation-Based Estimates Against the Experimental Dataset

The correlation-based estimates were validated against an experimental dataset obtained for pure HFE-7100 in the short multi-minichannel module (Setup #2, five parallel minichannels) under single-phase forced convection. The selected convection-only operating point corresponded to a mean mass flux of $G_{mean} = 372.5$ (kg/(m²·s)), inlet and outlet bulk-fluid temperatures of $T_{f,in} = 287.3$ K and $T_{f,out} = 288.5$ K, inlet and outlet gauge pressures of $p_{in} = 25,559.9$ Pa and $p_{out} = 24,399.2$ Pa, and an imposed heat flux of $q'' = 6473.2$ W/m² calculated from the measured electrical current and voltage. The IR-measured outer-foil temperature ranged from 294.7 K to 303.7 K; the ambient-air temperature was 294.9 K, and the atmospheric pressure was 0.96 bar.

The experimental heat transfer coefficient was evaluated from the spatial wall-temperature field measured by IR thermography. For each frame, the local coefficient was computed as

$$h_{exp}(x) = \frac{q''}{[T_F(x) - T_f(x) - \Delta T_{cond}]} \quad (6)$$

where $T_F(x)$ is the local outer-surface temperature of the heated foil measured by infrared thermography, $T_f(x)$ is the linear HFE-7100 bulk-temperature profile between the measured inlet/outlet temperatures (measured by K-type thermocouples placed in the inlet and outlet of the minichannel collector), and q'' is determined from the measured current and voltage, the heated area, and the assumed heat-loss correction.

Furthermore, ΔT_{cond} accounts for conduction through the heater foil and is calculated as follows:

$$\Delta T_{cond} = \frac{q'' \cdot \delta_F}{k_F} \quad (7)$$

where δ_F is the heater foil thickness and k_F is the foil conductivity.

A minimal quality criterion was applied (temperature difference in Equation (6) > 0.5 K and $h_{exp} > 0$), and $h_{exp,mean}$ was obtained by averaging over the retained axial positions. The correlation-based heat transfer coefficient h_{corr} was computed by first evaluating Re and Pr from Equations (1) and (2), then calculating Nu_{corr} using the laminar thermally developing correlation (3), and finally, converting Nu_{corr} to h_{corr} . For this first-stage validation, μ , c_p , and k were treated as constant reference values for HFE-7100 (near 298.1 K). For the selected 30 low-heat-flux frames (convection-only subset), the mean $h_{exp,mean}$ was 480.6 ± 11.7 W/(m²·K), while the mean h_{corr} was 555.2 ± 0.1 W/(m²·K). The mean relative difference was defined as

$$\Delta h_{rel} = \frac{(h_{exp} - h_{corr})}{h_{corr}} \cdot 100\%, \quad (8)$$

and equalled −13.44% (median −14.15%; 10th–90th percentiles −15.23% to −10.72%). The RMSE was 75.5 W/(m² K), and the MAPE (relative to h_{exp}) was 15.6%. This level of discrepancy is not negligible and confirms that the adopted correlation systematically overpredicts the absolute heat transfer coefficient for the validated case. The most likely contributors are the following: use of a circular-tube laminar entrance correlation with hydraulic diameter for a rectangular five-minichannel module, neglect of the Sieder–Tate viscosity correction, treatment of properties as spatially uniform, and omission of conjugate wall-conduction, and other module-specific effects from the reduced comparison model. For this reason, the framework is retained as a comparative screening tool under fixed assumptions, not as a bias-free predictive correlation for absolute heat transfer coefficient values.

Table 9 lists the 30 low-heat-flux (convection-only) validation cases for Setup #2, including the mass flux G , imposed heat flux q'' , the experimentally obtained mean heat transfer coefficient $h_{exp,mean}$, the corresponding correlation-based value h_{corr} , and the resulting relative difference Δh_{rel} .

Table 9. Validation dataset for the short module (Setup #2, HFE-7100): experimental mean heat transfer coefficient $h_{exp,mean}$, compared with the correlation-based value h_{corr} calculated using Equation (3), and the resulting relative difference Δh_{rel} calculated from Equation (8).

G (kg/(m ² ·s))	q'' (W/m ²)	$h_{exp,mean}$ (W/(m ² ·K))	h_{corr} (W/(m ² ·K))	Δh_{rel} (%)
372.2	4291.5	470.8	555.1	−15.2
372.1	4859.2	503.8	555.1	−9.2
372.2	5773.7	520.7	555.1	−6.2
372.7	6144.2	470.8	555.4	−15.2
372.4	6163.8	473.2	555.2	−14.8
372.4	6169.3	476.4	555.2	−14.2
372.1	6176.0	468.1	555.1	−15.7
372.3	6181.7	478.1	555.2	−13.9
372.1	6181.8	478.4	555.1	−13.8
372.4	6184.6	475.9	555.2	−14.3
372.5	6186.8	473.3	555.3	−14.8
372.3	6187.8	466.8	555.2	−15.9
372.6	6198.7	475.1	555.3	−14.4
372.5	6199.8	473.5	555.3	−14.7
372.5	6208.4	476.0	555.3	−14.3
372.5	6209.7	482.0	555.3	−13.2
372.6	6210.7	480.9	555.3	−13.4
372.3	6214.7	476.8	555.2	−14.1
372.4	6217.4	470.6	555.2	−15.2
372.4	6220.7	476.3	555.2	−14.2
372.3	6223.7	489.1	555.2	−11.9
372.6	6226.7	502.6	555.3	−9.5
372.4	6227.8	479.5	555.2	−13.6
372.0	6229.8	487.1	555.0	−12.2
372.4	6230.8	475.2	555.2	−14.4
372.4	6232.8	474.1	555.2	−14.6
372.7	6232.8	480.5	555.4	−13.5
372.6	6241.8	484.5	555.3	−12.8
372.6	6242.5	482.6	555.3	−13.1

372.5	6247.8	495.0	555.3	−10.9
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7. Conclusions

Thermophysical property data for three binary HFE/ethyl acetate systems (HFE-7100/EA, HFE-7300/EA, and HFE-73DE/EA) were compiled at 293.1, 313.1, and 328.1 K to support comparative single-phase forced-convection analysis in minichannel geometries. In the final assessment workflow, experimentally determined density, viscosity, and thermal conductivity were used directly, whereas mixture-specific heat capacity was estimated from pure-component mass fractions using a literature-supported ideal-mixture approximation solely for evaluation of the Prandtl number.

Under the adopted assumptions, the calculated correlation-based heat transfer coefficients span about 253–531 W/(m²·K) for the long reference module (Setup #1) and about 618–1296 W/(m²·K) for the short multi-minichannel module (Setup #2). For the HFE-7100/EA and HFE-7300/EA families, the highest h_{corr} values were obtained for the most EA-rich retained compositions, whereas within the retained HFE-73DE/EA subset, the 50/50 composition gave the highest h_{corr} because of its more favourable thermal conductivity level. In the corrected c_p treatment, the ranking is governed primarily by thermal conductivity, while c_p contributes more weakly through the one-third-power dependence embedded in the adopted correlation.

Validation against the short-module dataset for pure HFE-7100 showed a mean relative difference of −13.44% and a MAPE of 15.6%, confirming that the simplified framework systematically overpredicts the absolute heat transfer coefficient under the tested conditions. The most likely reasons are the use of a tube-derived entrance correlation with hydraulic diameter for a rectangular five-minichannel module, neglect of the viscosity-ratio correction, and omission of conjugate and module-specific effects. Accordingly, the proposed framework should be interpreted as a comparative screening tool under fixed assumptions rather than as a geometry-exact predictive model for absolute heat transfer coefficient values.

The dataset and workflow support early-stage selection of binary (and potentially more complex) dielectric working fluids for compact heat exchangers and mini-/micro-channel thermal management, where limited space, low fluid charge, and electrical safety are key constraints. Application areas include electronics and power electronics cooling, HVAC&R components with confined internal passages, and compact laboratory- or process-scale heat-exchanger modules requiring a balance between heat transfer performance, pressure-drop (pumping-power) penalties, and the target operating temperature range.

Limitations include the restriction to single-phase forced convection, the use of a tube-derived correlation applied via the hydraulic diameter, and the simplified treatment of thermophysical properties at each temperature level. Further work should extend validation to additional operating points and consider rectangular-duct-specific laminar formulations, temperature-dependent properties, and joint heat transfer/pressure-drop criteria for mixture down-selection.

Author Contributions: Conceptualization, A.P.; Methodology, A.P.; Software, M.P.; Validation, M.P.; Formal analysis, A.P. and M.P.; Investigation, A.P.; Resources, A.P.; Data curation, A.P. and M.P.; Writing—original draft, A.P. and M.P.; Writing—review & editing, A.P. and M.P.; Visualization, A.P.; Supervision, M.P.; Project administration, M.P.; Funding acquisition, M.P. All authors have read and agreed to the published version of the manuscript.

Funding: This research was funded by the National Science Centre, Poland, grant no. UMO-2025/57/B/ST8/00907. For the purpose of Open Access, the authors have applied a CC BY public copyright licence to the Author Accepted Manuscript (AAM) version arising from this submission.

Data Availability Statement: The original contributions presented in this study are included in the article. Further inquiries can be directed to the corresponding author.

Conflicts of Interest: The authors declare no conflicts of interest.

Nomenclature

Symbols

c_p	isobaric specific heat capacity ($\text{J}\cdot\text{kg}^{-1}\cdot\text{K}^{-1}$)
$c_{p,i}$	isobaric specific heat capacity of pure component i , ($\text{J}\cdot\text{kg}^{-1}\cdot\text{K}^{-1}$)
$c_{p,mix}$	estimated isobaric specific heat capacity of the mixture ($\text{J}\cdot\text{kg}^{-1}\cdot\text{K}^{-1}$)
D_h	hydraulic diameter, $D_h = 2ws/(w+s)$ (m)
G	mass flux ($\text{kg}\cdot\text{m}^{-2}\cdot\text{s}^{-1}$)
h_{exp}	experimental heat transfer coefficient, calculated according to Eq. (6) ($\text{W}\cdot\text{m}^{-2}\cdot\text{K}^{-1}$)
h_{corr}	correlation-based heat transfer coefficient ($\text{W}\cdot\text{m}^{-2}\cdot\text{K}^{-1}$)
$h_{exp,mean}$	experimentally obtained mean heat transfer coefficient ($\text{W}\cdot\text{m}^{-2}\cdot\text{K}^{-1}$)
I	electric current supplied to the heater (A)
k	thermal conductivity ($\text{W}\cdot\text{m}^{-1}\cdot\text{K}^{-1}$)
L	minichannel length (m)
Nu_{exp}	experimental Nusselt number, $Nu_{exp} = h_{exp} \cdot D_h / k$
Nu_{corr}	correlation-based Nusselt number (-)
p	pressure (Pa)
Pr	Prandtl number, $Pr = \mu \cdot c_p / k$ (-)
q''	heat flux at the heated wall ($\text{W}\cdot\text{m}^{-2}$)
Re	Reynolds number, $Re = G \cdot D_h / \mu$ (-)
s	minichannel depth (m)
T	temperature (K)
w	minichannel width (m)
w_i	mass fraction of component i in the mixture (-)
x	axial coordinate along the minichannel (m)

Greek letters

ΔT_{cond}	temperature drop across the heater foil due to one-dimensional conduction $\Delta T_{cond} = q'' \cdot \delta / k_f$ (K)
Δh_{rel}	relative difference between $h_{exp,mean}$ and h_{corr} (%)
μ	dynamic viscosity ($\text{Pa}\cdot\text{s}$)
μ_w	dynamic viscosity evaluated at the wall temperature ($\text{Pa}\cdot\text{s}$)
ν	kinematic viscosity ($\text{m}^2\cdot\text{s}^{-1}$)
ρ	density ($\text{kg}\cdot\text{m}^{-3}$)
δ	heater foil thickness (m)

Subscripts

$corr$	value calculated from a correlation
exp	value obtained from experiment
F	foil (heater wall)
f	fluid (bulk)
i	component index
in/out	inlet/outlet
max	maximum value
$mean$	mean (averaged) value
min	minimum value
mix	mixture

Abbreviations

EA	ethyl acetate
HFE	hydrofluoroether
IR	infrared thermography
LCT	liquid crystal thermography

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